

FedDSG: Backdoor Defense via Semantic Filter and Geometric Constraint in Federated Learning

Abstract

Backdoor attacks pose a serious threat to Internet-of-Things (IoT) federated learning. In IoT deployments, pronounced non-independent and identically distributed (non-IID) data heterogeneity causes benign client updates to exhibit substantial variability across devices. Meanwhile, the physical exposure of IoT devices increases the risk of large-scale compromise and elevated malicious participation. Such variability allows poisoned updates to blend into natural fluctuations, rendering many robust aggregation and detection-based defenses unreliable. We propose *FedDSG*, a server-side defense that combines a semantic bias filter and a geometric direction constraint to counter backdoor manipulation. FedDSG first extracts a novel scale-invariant semantic cue from the last-layer bias of client updates to identify abnormal target-class reinforcement, staying effective even when benign bias patterns differ substantially across clients. The remaining updates are then constrained using a reference derived from a small trusted anchor set, limiting adversarial drift. This sequential design links semantic cues with geometric structure, where the former removes clearly suspicious updates and the latter stabilizes the residual ones, preventing misdetection-induced drift amplification while avoiding distortion of benign updates. The method does not alter client behavior or communication and adds minimal server-side overhead. Extensive experiments on MNIST, Fashion-MNIST, CIFAR-10, and SVHN under non-IID distributions with high malicious participation demonstrate the robustness of FedDSG. It reduces the attack success rate to 0.003, 0.006, 0.007, and 0.091, respectively, with only marginal accuracy loss and consistently achieves the highest Overall Performance Score (OPS), reflecting a superior trade-off between robustness and accuracy. Code and data availability information is provided in the *Availability of data and materials* section.

Keywords: Federated learning, IoT security, Backdoor attacks, non-IID heterogeneity

047 1 Introduction

048
049 Federated learning (FL) has become a practical paradigm for training models over
050 large populations of Internet-of-Things (IoT) devices [1]. By keeping raw data
051 on-device, FL enables privacy-preserving analytics for environmental monitoring,
052 industrial automation, and urban sensing. Yet, cross-device IoT federations are
053 inherently heterogeneous: sensing contexts vary across locations, participation is
054 sporadic, and device resources are uneven [2–4]. These factors induce pronounced non-
055 independent and identically distributed (non-IID) data distributions, so even benign
056 clients may produce updates that differ substantially in both semantics and geometry.
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058 This heterogeneity is particularly problematic when the system faces backdoor
059 attacks. Many IoT devices are physically exposed and operate with limited security
060 hardening, making persistent data poisoning realistic [5, 6]. In heterogeneous regimes,
061 benign updates naturally exhibit high variability, and an attacker can exploit this
062 variability to blend in while gradually biasing the global model toward a trigger-
063 dependent decision rule. The overall attack process is illustrated in Fig. 1.
064

065 **Challenge.** Cross-device IoT federations expose two fundamental obstacles for
066 backdoor defense:
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- 068 • *Non-IID heterogeneity undermines proximity-based detection.* Diverse sensing con-
069 texts, hardware conditions, and participation patterns induce large variability in
070 both the semantics and geometry of benign updates, allowing malicious updates
071 to blend into this natural dispersion and evade distance-based filtering, clustering,
072 and anomaly detection [7, 8].
- 073 • *Benign-majority assumptions fail under high malicious participation.* Byzantine-
074 robust aggregation rules [9–11] require benign-majority or IID-like concentration,
075 yet IoT deployments often involve high malicious ratios and persistent backdoor
076 strategies that violate these assumptions.
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Existing federated backdoor defenses fall into three categories: pure robust aggregation, anomaly detection, and hybrid approaches. Robust aggregation suppresses extreme updates through statistical or geometric selection but relies on benign-majority and concentration assumptions that fail under severe non-IID heterogeneity. Detection-based methods identify suspicious clients via similarity metrics, clustering, or structural patterns in update space [8, 12]. Semantic and layer-level strategies exploit cues from specific model components rather than full-model geometry. However, these approaches become unreliable when benign variability overlaps with adversarial mimicry. Hybrid methods combine detection with robust aggregation for more rigorous defense logic and theoretically stronger robustness. However, they incur higher computational costs and remain vulnerable under pronounced data heterogeneity with high malicious ratios despite their combined strengths.

Our Approach. We propose *FedDSG*, a backdoor *Defense* via a *Semantic* bias filter and a *Geometric* direction constraint in *Federated* Learning. This dual-component design addresses a fundamental limitation of existing approaches: purely robust aggregation discards informative semantic cues for distinguishing benign updates, while purely anomaly detection becomes unstable under non-IID fluctuations. FedDSG resolves this tension by leveraging a structural asymmetry in bias updates. Targeted backdoor optimization systematically drives final-layer bias parameters toward a specific target class, whereas benign updates, despite substantial dispersion, retain a coarse task-aligned component that can be approximated using a small server-side anchor dataset. Building on this distinction, FedDSG combines detection and correction mechanisms. SBF identifies and removes highly suspicious clients using scale-invariant semantic scores derived from bias updates. GDC stabilizes the remaining updates by constraining their geometric deviation from a benign reference direction estimated from the anchor set. This design reduces attack success rates while preserving the natural diversity of benign updates in heterogeneous IoT federations.

Contributions. The main contributions are as follows.

- We propose FedDSG, a dual-component defense that combines a semantic bias filter (SBF) with a geometric direction constraint (GDC). SBF detects abnormal reinforcement toward a specific target class from bias updates, while GDC regulates residual drift using a benign reference direction. Together, these components suppress targeted backdoor influence while preserving the natural variability of benign updates.
- We introduce a scale-invariant semantic cue that measures abnormal amplification toward the target class in bias updates. The cue remains stable under heterogeneous local training dynamics and enables reliable identification of persistent backdoor behavior.
- We conduct extensive evaluations on MNIST, Fashion-MNIST, CIFAR-10, and SVHN under both IID and non-IID settings. FedDSG maintains competitive clean accuracy and substantially reduces attack success rates across all datasets, demonstrating stable performance under heterogeneous IoT conditions.

Paper Organization. Section 2 reviews related work. Section 3 introduces the system and threat model. Section 4 details the FedDSG framework. Section 5 provides the convergence analysis, including the bounded-bias descent characterization and the resulting stationary-point guarantee. Section 6 reports experimental results. Section 7 discusses limitations, and Section 8 concludes the paper.

2 Related Work

This section reviews related work on federated backdoor attacks and defenses. We summarize representative attack methods and threat models, followed by existing defense approaches and their limitations in cross-device IoT settings.

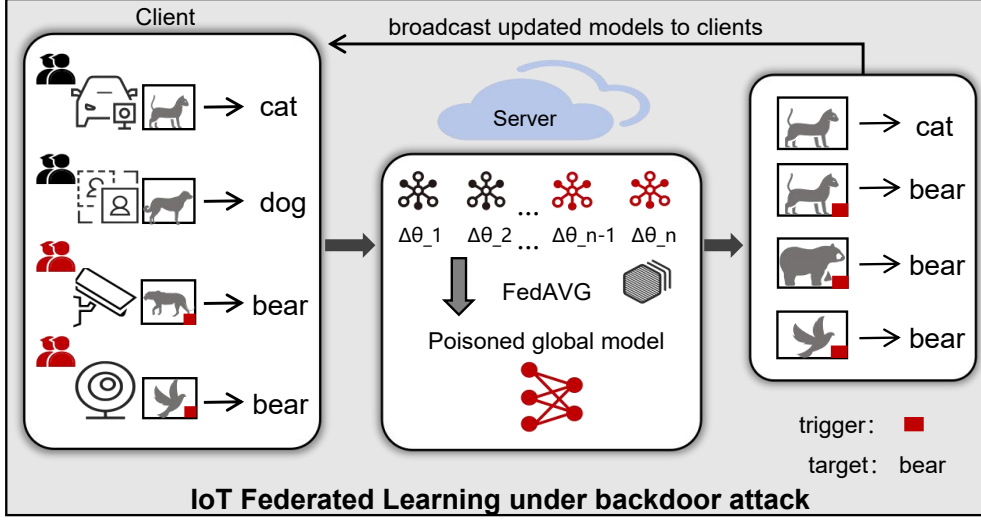


Fig. 1 IoT backdoor attack. Malicious clients inject trigger-patched samples during local training, causing the server to aggregate poisoned updates and the global model to misclassify triggered inputs into the target class.

2.1 Federated Backdoor Attacks

Backdoor attacks implant trigger-based misbehavior while maintaining high clean accuracy [13]. Early attacks such as BadNets use visible pixel patterns as triggers [14]. Model-replacement attacks send large, crafted updates to dominate aggregation. Distributed backdoor attacks like DBA [15] split the trigger across multiple compromised clients for stealth. Continuous poisoning methods [16] iteratively adapt malicious updates to remain plausible under heterogeneous training.

Backdoor attacks can also evade or manipulate robust aggregation rules [17], since these methods rely on benign updates forming a compact cluster that often does not exist under highly dispersed IoT data. Recent work studies stronger threat models, including personalization-aware backdoors, multi-label objectives, and non-cooperative adversaries [18–21], often in resource-constrained IoT scenarios. IoT-focused FL security studies [22, 23] highlight the need for defenses that remain

effective under the complex, highly heterogeneous data distributions typical of IoT networks [24, 25].

2.2 Federated Backdoor Defenses

Existing federated backdoor defenses can be broadly organized into three categories: robust aggregation, anomaly detection, and hybrid integration. Table 1 summarizes representative methods and their key characteristics.

Robust Aggregation. Robust aggregation methods perform correction by modifying the aggregation rule. Krum and Multi-Krum [9] select updates based on distance, choosing those closest to the majority in Euclidean space. Median and Trimmed-Mean [10] apply coordinate wise statistics, using parameter medians or trimmed averages to suppress extreme values. Bulyan [11] further strengthens this strategy through multi-stage trimming. FLTrust [26] introduces cosine weighted aggregation guided by a trusted server reference gradient. These correction oriented approaches implicitly assume that benign updates exhibit sufficient geometric consistency. However, IoT heterogeneity violates this assumption due to diverse sensing conditions and optimization behaviors [27, 28]. Even enhanced variants [29–31] struggle under large benign variance and persistent adversaries [32].

Anomaly Detection. Detection methods identify suspicious clients before aggregation by analyzing structural patterns in updates. FoolsGold [33] performs similarity based downweighting to mitigate collusive behavior among clients with highly aligned updates. FLAB [34] detects bias anomalies by analyzing abnormal variations in the last layer bias. FLAME [8] applies clustering followed by update clipping to isolate and suppress anomalous groups. These approaches rely on stable similarity relationships or separable update structures, which IoT heterogeneity disrupts through semantic inconsistencies and drifting clusters [35].

Hybrid Integration. Hybrid defenses combine detection with robust aggregation. FedDLAD [12] classifies clients by analyzing parameter connectivity and performs secondary correction across update dimensions to identify and flip anomalies. However, these methods still depend on stable references and clustering structures that collapse under strong non-IID distributions and high malicious ratios [32].

Summary. While effective under moderate conditions, existing defenses fail in cross-device IoT settings where strong non-IID heterogeneity breaks geometric concentration, disrupts similarity structures, and destabilizes semantic signals [27, 28, 32, 35, 36]. These failures stem from collapsed core assumptions under pronounced heterogeneity and adversarial dominance. Motivated by these limitations, we propose FedDSG, combining a semantic filter with geometric constraint correction to maintain effectiveness under realistic IoT non-IID and high-adversary regimes.

3 Problem Setting and Threat Model

This section defines the federated learning setting and the threat model, including the server assumptions, attacker capabilities, and the cross-device IoT scenario considered in this study.

3.1 Federated Learning Setting

We consider a standard cross-device federated learning paradigm, as typically adopted in large-scale IoT deployments, in which a global model is collaboratively trained across a population of heterogeneous edge devices. In each communication round t , the central server samples a subset of participating clients $\mathcal{S}_t$, broadcasts the current global model w^t , and updates it based on an aggregation of the clients' local updates:

$$w^{t+1} = w^t + \eta \sum_{k \in \mathcal{S}_t} \lambda_k u_k^t, \quad (1)$$

Table 1 Defense comparison.

Method	Mechanism	Complexity	Detection	Correction	Anchor-free	Non-IID	High-Malicious
Krum / Multi-Krum	Distance selection	$O(N^2d)$	×	✓	✓	×	×
Median	Coordinate median	$O(dN \log N)$	×	✓	✓	×	×
Trimmed-Mean	Parameter trimming	$O(dN \log N)$	×	✓	✓	×	×
Bulyan	Multi-stage trimming	$O(N^2d)$	×	✓	✓	×	×
FLTrust	Cosine-weighted aggregation	$O(Nd)$	×	✓	×	✓	✓
FoolsGold	Similarity downweighting	$O(N^2d)$	✓	×	✓	×	×
FLAB	Bias anomaly detect.	$O(NK)$	✓	×	✓	×	✓
FLAME	Clustering + clipping	$O(N^2d)$	✓	×	✓	✓	×
FedDLAD	Connectivity analysis + flipping	$O(N^2d)$	✓	✓	✓	✓	×
FedDSG (Ours)	Bias filter + Dir.constraint	O(NK)	✓	✓	×	✓	✓

N denotes the number of participating clients per round, d the model dimension, and K the number of classes. Methods are grouped as robust aggregation, anomaly detection, and hybrid integration.

where u_k^t denotes the model update computed using client k 's private dataset D_k , λ_k is the aggregation weight associated with client k , and η is the server-side learning rate.

In realistic IoT scenarios, clients exhibit heterogeneous sensing modalities, highly imbalanced label distributions, and diverse environmental conditions. This intrinsic heterogeneity induces substantial variability in local training dynamics and leads to pronounced semantic inconsistency and geometric drift among benign client updates. Such phenomena complicate the reliable discrimination between naturally occurring benign variability and adversarially crafted manipulations.

3.2 Threat Model

We consider cross-device IoT federated learning under heterogeneous non-IID data distributions, where benign client updates may vary substantially in both magnitude and direction. Such dispersion makes it difficult to distinguish natural variability from adversarial manipulation.

Attacker. An adversary compromises a subset of participating devices, potentially at a high malicious ratio. A malicious device is fully controlled by the attacker and may poison local data, relabel samples, execute arbitrary local training, and post-process model updates before uploading them. The attacker cannot tamper with benign clients, the communication channel, or the server aggregation mechanism. We do not assume that the attacker has exact knowledge of the internal filtering and correction rules deployed by the server. Therefore, adaptive strategies that intentionally hide the backdoor from the monitored bias subspace, such as freezing the last-layer bias and shifting the malicious effect to weight parameters, are outside the main threat model.

Compromised devices inject a fixed trigger into a fraction of local samples and relabel them to a designated target class. When selected, malicious devices upload updates

computed on poisoned data, enabling persistent influence across rounds. The attack objective is to preserve normal predictions on clean inputs while forcing triggered inputs to a fixed target label:

$$\mathcal{M}_w(x) = y, \quad \mathcal{M}_w(x') = y_{\text{target}}. \quad (2)$$

Each malicious client optimizes the following local objective:

$$\begin{aligned} \mathcal{L}_u = & \mathbb{E}_{(x,y) \sim D_u} \ell(f_w(x), y) \\ & + \lambda \mathbb{E}_{(x,y) \sim D_u^{\text{poison}}} \ell(f_w(\tau(x)), y_{\text{target}}). \end{aligned} \quad (3)$$

Here $\tau(\cdot)$ denotes trigger injection and λ controls the attack strength.

Defender. The server is assumed to be honest but does not trust participating clients. It observes only uploaded model updates and does not access raw local data. The aggregation protocol remains unchanged. The server may maintain a small clean anchor set, independent of participating clients’ private data, to estimate a benign reference direction. Because this reference direction is used as a coarse geometric constraint rather than an exact representation of the global data distribution, the anchor set does not need to be strictly class-balanced. Slightly non-IID or partially incomplete clean anchor data are easier to obtain in practice and remain within a reasonable defender capability assumption.

4 Methodology

This section presents the FedDSG framework for federated backdoor defense. It comprises two complementary modules: a scale-invariant semantic bias filter (SBF) to detect concentrated target reinforcement, and a geometric direction constraint (GDC) to regulate directional deviation with a clean anchor reference. We also describe the

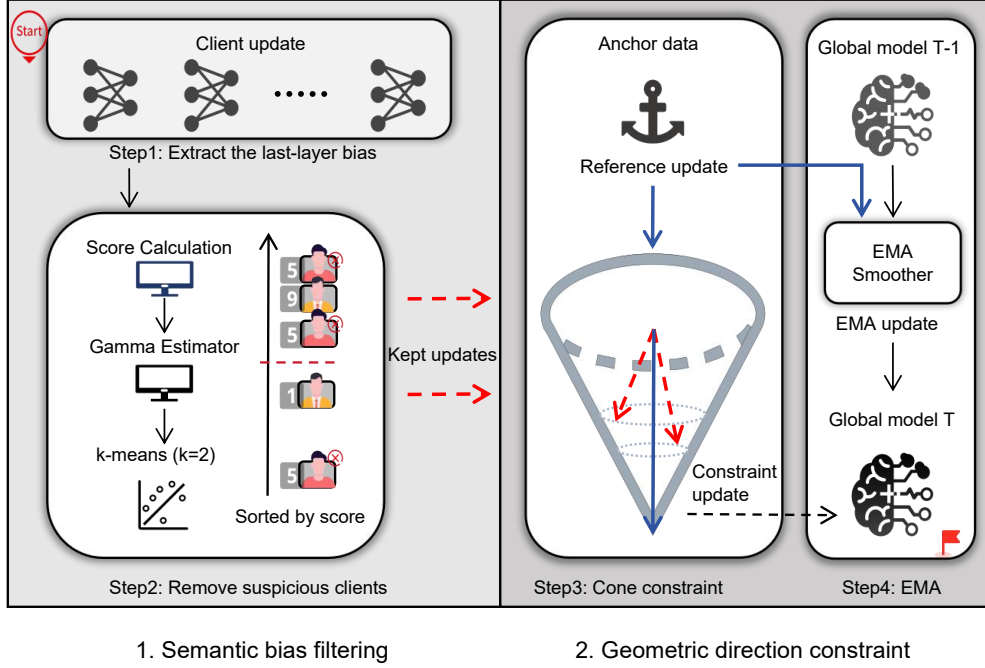


Fig. 2 FedDSG pipeline. Step 1 extracts the last-layer bias updates. Step 2 performs SBF-based detection and removes suspicious clients. Step 3 applies GDC to constrain direction using a clean anchor reference. Step 4 updates the global model with EMA smoothing.

full pipeline and analyze its computational complexity and operational properties in heterogeneous IoT settings.

4.1 Overview

FedDSG is motivated by the difficulty of defending against backdoor attacks in cross device IoT federated learning under heterogeneous local training dynamics and high malicious participation ratios. Under such conditions, detection-based defenses may discard benign updates due to natural dispersion, while pure robust aggregation rules may allow consistent reinforcement toward a target class from multiple malicious clients to accumulate in the aggregated model.

FedDSG combines detection and correction at the bias level to address these complementary weaknesses. The Semantic Bias Filter (SBF) performs detection by

identifying clients whose bias updates exhibit consistent reinforcement toward a target class using scale normalized semantic cues. This step removes concentrated malicious amplification while preserving retained benign updates.

The Geometric Direction Constraint (GDC) performs correction by regulating the remaining updates with respect to a reference direction estimated from a small clean anchor set maintained at the server. This constraint limits the accumulation of residual directional drift across rounds, even when some malicious updates are not removed.

SBF controls explicit target reinforcement, and GDC stabilizes aggregation against gradual deviation. Together they provide selective removal and bounded correction under non-IID heterogeneity and elevated malicious participation. An overview of the complete pipeline is illustrated in Fig. 2.

4.2 SBF: Semantic Bias Filter

For a K -class classifier, the last-layer bias update $\Delta b_s^t \in \mathbb{R}^K$ records how client s adjusts class logits during local training in round t . Here, K is the number of classes, and the class index is denoted by $c \in \mathcal{C}$ with $\mathcal{C} = \{1, \dots, K\}$. Because this bias block directly shifts class logits, it is tightly aligned with decision behavior and provides a compact class-level view of local optimization dynamics.

However, under heterogeneous IoT conditions, the raw bias update alone is not sufficient for distinguishing benign and malicious clients. Benign updates often exhibit mixed and device-specific fluctuations across classes due to data imbalance and local optimization variability. At the same time, full-parameter distances are unreliable because high dimensionality and heterogeneous training trajectories obscure class-level effects. Therefore, extracting meaningful adversarial signals requires an appropriate transformation that separates persistent target reinforcement from benign variation.

Targeted poisoning induces a structurally different pattern. Repeated steering toward a fixed target class produces a consistent extremal tendency in the bias coordinate associated with that target, even when the remaining components of the update vary with local data. This asymmetry makes the bias block a natural entry point for identifying adversarial tendencies once scale and offset effects are properly controlled. FedDSG therefore does not rely on the raw bias values.

SBF extracts semantic evidence of targeted reinforcement by analyzing the final-layer bias update, a parameter block whose dimensionality and semantics make it more robust to non-IID variability than full-parameter geometry. Let $\Delta b_s^t \in \mathbb{R}^K$ denote the last-layer bias update uploaded by client s in round t . Let steps_s be the effective number of local optimization steps executed by client s in that round. Concretely, if the client runs E local epochs with batch size B_s over a local dataset of size $|D_s|$, one typical choice is $\text{steps}_s = E \cdot \lceil |D_s|/B_s \rceil$; more generally, steps_s can be set to the exact number of SGD iterations performed.

Because clients differ in local batch sizes and iteration counts, raw update magnitudes are not directly comparable. We therefore start with step normalization:

$$\tilde{b}_s^t[c] = \frac{\Delta b_s^t[c]}{\text{steps}_s}, \quad (4)$$

where $\tilde{b}_s^t \in \mathbb{R}^K$ is the step-normalized bias update and $\tilde{b}_s^t[c]$ denotes its c -th coordinate.

Under targeted backdoor poisoning, the attacker consistently reinforces the same target class, producing a recurring extremum in the normalized bias update. We capture this phenomenon by identifying the coordinate exhibiting the strongest reinforcing effect:

$$\gamma_s = \arg \max_{c \in \mathcal{C}} \tilde{b}_s^t[c], \quad (5)$$

where $\gamma_s \in \mathcal{C}$ is the dominant class index under our sign convention.

599 Benign updates may contain offsets caused by non-uniform class exposure. To
600 expose relative class-selective deviations, we compute a robust center from all non-
601 dominant coordinates:
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$$603 \quad m_s = \text{median}\{\tilde{b}_s^t[c] : c \in \mathcal{C}, c \neq \gamma_s\}, \quad (6)$$

604
605 where the median is taken over the $(K - 1)$ scalar entries excluding the dominant
606 coordinate. We then subtract it to obtain
607

$$608 \quad d_s[c] = \tilde{b}_s^t[c] - m_s, \quad (7)$$

609 where $d_s \in \mathbb{R}^K$ denotes the centered deviation vector.
610

611 We convert centered deviations into nonnegative energies:

$$612 \quad E_s[c] = d_s[c]^2, \quad (8)$$

613
614 where $E_s \in \mathbb{R}_{\geq 0}^K$. Here, $E_s[c]$ captures the magnitude of class-wise semantic drift in a
615 scale-invariant manner, forming a class-level “energy profile” of client s . Let $\|E_s\|_1 =$
616 $\sum_{c \in \mathcal{C}} E_s[c]$ denote the ℓ_1 norm (sum of energies).
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618 To characterize the sharpness of this profile, we define a concentration ratio

$$619 \quad \text{conc}_s = \frac{E_s[\gamma_s]}{\sum_{c \in \mathcal{C}} E_s[c]}, \quad (9)$$

620
621 which measures the proportion of total energy contributed by the previously identified
622 dominant class γ_s . A high conc_s indicates that the client’s bias pattern exhibits a *single*
623 *pronounced peak*, a signature often associated with class-targeted steering behaviors
624 in backdoor updates.
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We further capture the contrast between the dominant class and all remaining classes via a dominance ratio

$$\text{str}_s = \frac{E_s[\gamma_s]}{\sum_{c \in \mathcal{C}, c \neq \gamma_s} E_s[c] + \varepsilon}, \quad (10)$$

where $\varepsilon > 0$ is a small constant added to avoid division by zero. While conc_s reflects the *global concentration* of the energy, str_s measures the *relative dominance* of the peak over the residual background. Large str_s values therefore signify disproportionate amplification toward a specific class, providing a strong cue for identifying target-consistent bias under backdoor attacks.

We combine them into a single scale-invariant peak-bias score:

$$\text{score}_s = \text{conc}_s \cdot \log(1 + \text{str}_s), \quad (11)$$

where the logarithmic term prevents extreme cases from dominating the statistic while preserving monotonicity in str_s .

We then cluster the scalar scores $\{\text{score}_s\}_{s \in \mathcal{S}_t}$ using 1D k -means with $k = 2$. Let $\mathcal{S}_{\text{susp}}$ denote the cluster with the larger mean score (the suspicious group). The target class is inferred through score-weighted voting:

$$\hat{y} = \arg \max_{c \in \mathcal{C}} \sum_{s \in \mathcal{S}_{\text{susp}}} \mathbb{I}(\gamma_s = c) \text{score}_s, \quad (12)$$

where $\mathbb{I}(\cdot)$ is the indicator function and $\hat{y} \in \mathcal{C}$ is the estimated target class. Finally, all clients with $\gamma_s = \hat{y}$ are discarded. Algorithm 1 provides the implementation details.

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691 Algorithm 1 SBF
692   Input: Sampled clients  $S_t$ ; updates  $\{u_s^t\}$ ; steps  $\{\text{steps}_s\}$ ; class set  $\mathcal{C}$ ;  $\varepsilon > 0$ 
693   Output: Kept clients  $S_{\text{keep}}$ ; estimated target  $\hat{y}$ 
694   1: for all  $s \in S_t$  do
695     2:  $\Delta b_s^t \leftarrow \text{EXTRACTBIAS}(u_s^t)$ 
696     3:  $\tilde{b}_s^t \leftarrow \Delta b_s^t / \text{steps}_s$ 
697     4:  $\gamma_s \leftarrow \arg \max_{c \in \mathcal{C}} \tilde{b}_s^t[c]$ 
698     5:  $m_s \leftarrow \text{median}\{\tilde{b}_s^t[c] : c \in \mathcal{C}, c \neq \gamma_s\}$ 
699     6:  $d_s \leftarrow \tilde{b}_s^t - m_s$ ;  $E_s \leftarrow d_s \odot d_s$ 
700     7:  $\text{conc}_s \leftarrow E_s[\gamma_s] / \|E_s\|_1$ 
701     8:  $\text{str}_s \leftarrow E_s[\gamma_s] / (\|E_s\|_1 - E_s[\gamma_s] + \varepsilon)$ 
702     9:  $\text{score}_s \leftarrow \text{conc}_s \cdot \log(1 + \text{str}_s)$ 
703   10: end for
704   11: Run 1D  $k$ -means ( $k = 2$ ) on  $\{\text{score}_s\}$ ; let  $\mathcal{S}_{\text{susp}}$  be the higher-mean cluster
705   12:  $V[c] \leftarrow 0$  for all  $c \in \mathcal{C}$ 
706   13: for all  $s \in \mathcal{S}_{\text{susp}}$  do
707     14:  $V[\gamma_s] \leftarrow V[\gamma_s] + \text{score}_s$ 
708   15: end for
709   16:  $\hat{y} \leftarrow \arg \max_{c \in \mathcal{C}} V[c]$ 
710   17:  $S_{\text{keep}} \leftarrow \{s \in S_t : \gamma_s \neq \hat{y}\}$ 
711   18: return  $S_{\text{keep}}, \hat{y}$ 

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4.3 GDC: Geometric Direction Constraint

While SBF removes clients exhibiting high-salience target reinforcement, low-salience adversarial drift may still remain and accumulate over rounds. Such drift is difficult to detect from per-client statistics alone because heterogeneous IoT clients naturally produce diverse updates. GDC complements SBF by introducing a trusted benign reference direction derived from a small anchor dataset maintained at the server, enabling the server to constrain residual deviations without relying on a benign-majority assumption.

Benign reference construction. Given a small trusted anchor dataset $\mathcal{D}_a$, drawn from clean calibration data or other trusted sources, we compute the bias gradient induced by clean data:

$$r_b^t = -\nabla_b \mathcal{L}(\theta^{t-1}; \mathcal{D}_a), \quad (13)$$

where $\mathcal{L}(\cdot)$ is the standard training loss (e.g., cross-entropy), and ∇_b denotes the gradient with respect to the last-layer bias parameters b .

Since the anchor dataset is typically small, the gradient direction may fluctuate across rounds due to sampling noise and model evolution. Using the raw reference r_b^t directly may therefore introduce instability in the geometric constraint. To obtain a more stable estimate of the trusted descent direction, we maintain an exponential moving average (EMA):

$$\hat{r}_b^t = (1 - \mu)\hat{r}_b^{t-1} + \mu r_b^t, \quad (14)$$

where $\mu \in (0, 1)$ controls the smoothing factor and $\hat{r}_b^t$ serves as the stabilized reference vector in round t . In practice, we can compute the full gradient $-\nabla_\theta \mathcal{L}(\theta^{t-1}; \mathcal{D}_a)$ and then extract its last-layer bias block.

The reference vector $\hat{r}_b^t$ represents the descent direction induced by trusted clean data. Updates aligned with this direction are consistent with minimizing the clean loss defined on the anchor set. In contrast, components that deviate substantially from this direction alter the bias parameters in ways that are not supported by the trusted objective. Under persistent adversarial influence, such deviations may accumulate across rounds and gradually distort the decision boundary, even if their magnitude is moderate in each individual round. This motivates separating each client bias update into a reference-aligned component and an orthogonal residual, and constraining only the deviation from the trusted direction.

Geometric decomposition. For each surviving client, GDC analyzes how its bias update aligns with the reference. Let Δb_s^t be the last-layer bias update of client s in round t . We decompose it into a reference-parallel component and an orthogonal residual:

$$b_s^\parallel = \frac{\langle \Delta b_s^t, \hat{r}_b^t \rangle}{\|\hat{r}_b^t\|^2 + \varepsilon} \hat{r}_b^t, \quad (15)$$

$$b_s^\perp = \Delta b_s^t - b_s^\parallel, \quad (16)$$

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783 Algorithm 2 GDC
784 Input: Global model  $\theta^{t-1}$ ; kept updates  $\{u_s^t\}_{s \in S_{\text{keep}}}$ ; anchor set  $\mathcal{D}_a$ ; weights  $\{\lambda_s\}$ ;
785 EMA factor  $\mu$ ; cone cap  $c_{\text{ortho}}$ ;  $\varepsilon > 0$ 
786 Output: Aggregated update  $\Delta^t$ 
787 1:  $g \leftarrow -\nabla_{\theta} \mathcal{L}(\theta^{t-1}; \mathcal{D}_a)$ 
788 2:  $r_b^t \leftarrow \text{EXTRACTBIAS}(g)$ 
789 3:  $\hat{r}_b^t \leftarrow (1 - \mu)\hat{r}_b^{t-1} + \mu r_b^t$ 
790 4: for all  $s \in S_{\text{keep}}$  do
791 5:    $\Delta b_s^t \leftarrow \text{EXTRACTBIAS}(u_s^t)$ 
792 6:    $b_s^{\parallel} \leftarrow \frac{\langle \Delta b_s^t, \hat{r}_b^t \rangle}{\|\hat{r}_b^t\|^2 + \varepsilon} \hat{r}_b^t$ 
793 7:    $b_s^{\perp} \leftarrow \Delta b_s^t - b_s^{\parallel}$ 
794 8:   if  $\|b_s^{\perp}\| > c_{\text{ortho}} \|b_s^{\parallel}\|$  then
795 9:      $b_s^{\perp} \leftarrow \frac{c_{\text{ortho}} \|b_s^{\parallel}\|}{\|b_s^{\perp}\| + \varepsilon} b_s^{\perp}$ 
796 10:  end if
797 11:   $\bar{b}_s \leftarrow b_s^{\parallel} + b_s^{\perp}$ 
798 12:   $\bar{u}_s^t \leftarrow \text{REPLACEBIAS}(u_s^t, \bar{b}_s)$ 
799 13: end for
800 14:  $\Delta^t \leftarrow \sum_{s \in S_{\text{keep}}} \lambda_s \bar{u}_s^t$ 
801 15: return  $\Delta^t$ 

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where $\langle \cdot, \cdot \rangle$ and $\|\cdot\|$ denote the Euclidean inner product and ℓ_2 norm, respectively, and $\varepsilon > 0$ avoids division by zero.

Soft deviation control. Rather than discarding updates with large orthogonal components, GDC applies a mild cone constraint:

$$\|b_s^{\perp}\| \leq c_{\text{ortho}} \|b_s^{\parallel}\|, \quad (17)$$

where $c_{\text{ortho}} > 0$ controls the maximum allowed ratio between the orthogonal component and the reference-parallel component. When the bound is violated, we rescale only the orthogonal component to satisfy Eq. (17). The adjusted bias update is then reinserted into the full client update so that the geometric constraint is applied only to the bias block.

Algorithm 2 gives the implementation details.

4.4 Computational Complexity

FedDSG targets large-scale IoT FL settings in which server-side overhead is a primary constraint. Client overhead is zero because local training and upload formats remain unchanged. Per round, the server extracts the last-layer bias-update blocks in $O(|S_t|K)$, where $|S_t|$ is the number of sampled clients and K is the number of classes. For SBF, computing each client's score involves step normalization (Eq. (4)), a robust offset via the median over $K-1$ entries (Eq. (6)), and element-wise energy statistics (Eqs. (8)–(11)), costing $O(K)$ with a selection-based median (or $O(K \log K)$ if implemented via sorting). Clustering operates on scalar scores and costs $O(|S_t|)$ for a small number of iterations. GDC performs K -dimensional projections and (optional) rescaling (Eqs. (15)–(17)), costing $O(|S_t|K)$. Overall, the per-round server overhead is $O(|S_t|K)$ in practice, since K is typically small (e.g., 10–100), all operations are confined to the bias-update block, and communication remains identical to FedAvg.

5 Convergence Analysis

We analyze the convergence properties of FedDSG under standard assumptions for non-convex stochastic federated optimization. The server update rule is

$$\begin{aligned} w^{t+1} &= w^t + \eta \sum_{s \in S_t^{\text{keep}}} \lambda_s \bar{u}_s^t \\ &\triangleq w^t + \eta \Delta^t, \end{aligned} \tag{18}$$

where Δ^t denotes the aggregated update, $\lambda_s \geq 0$, and $\sum_{s \in S_t^{\text{keep}}} \lambda_s = 1$.

5.1 Assumptions

Let

$$F(w) = \sum_{k=1}^K p_k F_k(w)$$

875 be the global objective, where p_k denotes the relative weight of client k in the
 876 population.
 877

878 **A1 (Smoothness).** Each F_k is L -smooth, i.e.,
 879

$$\|\nabla F_k(u) - \nabla F_k(v)\| \leq L\|u - v\|.$$

880
 881
 882 This condition ensures that the local objectives admit quadratic upper bounds.
 883

884
 885 **A2 (Bounded variance).** For stochastic gradients $g_k(w; \xi)$,
 886

$$\mathbb{E}_\xi[g_k(w; \xi)] = \nabla F_k(w), \tag{19}$$

$$\mathbb{E}_\xi\|g_k(w; \xi) - \nabla F_k(w)\|^2 \leq \sigma^2. \tag{20}$$

887
 888
 889
 890 This bounds stochastic noise from local minibatch sampling.
 891

892 **A3 (Client heterogeneity).**
 893

$$\mathbb{E}_k\|\nabla F_k(w) - \nabla F(w)\|^2 \leq \zeta^2.$$

894
 895
 896 The constant ζ^2 measures statistical heterogeneity across clients and vanishes under
 897 IID data.
 898

899
 900 **A4 (Random participation).** The sampled set S_t is drawn independently of local
 901 minibatch randomness conditioned on w^t . This avoids coupling between participation
 902 and gradient noise.
 903

904 **A5 (SBF filtering event).** There exists $\delta \in [0, 1)$ such that with probability at
 905 least $1 - \delta$, the kept set satisfies: (i) benign weight $\geq \rho_b$; (ii) misfiltered benign weight
 906 $\leq \epsilon_b$; (iii) retained malicious weight $\leq \epsilon_m$. This assumption formalizes the statistical
 907 reliability of SBF screening.
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A6 (Bounded updates). There exists $G > 0$ such that

$$\|\bar{u}_s^t\| \leq G \quad \forall s, t.$$

This ensures that the aggregated update remains uniformly bounded.

5.2 Convergence Guarantee

We first establish two auxiliary lemmas that characterize the geometric effect of GDC and the bias introduced by SBF filtering.

Lemma 1 (Boundedness of GDC) *For any x ,*

$$\|T_t(x)\| \leq (1 + c_{\text{ortho}})\|x\|.$$

Proof Write $x = x^{\parallel} + x^{\perp}$. If $\|x^{\perp}\| \leq c_{\text{ortho}}\|x^{\parallel}\|$, then $T_t(x) = x$. Otherwise, GDC rescales $x^{\perp}$ so that

$$\|T_t(x)^{\perp}\| = c_{\text{ortho}}\|x^{\parallel}\|.$$

$$\begin{aligned} \|T_t(x)\| &\leq \|x^{\parallel}\| + \|T_t(x)^{\perp}\| \\ &= (1 + c_{\text{ortho}})\|x^{\parallel}\| \\ &\leq (1 + c_{\text{ortho}})\|x\|. \end{aligned} \tag{21}$$

□

Lemma 2 (Approximate descent) *There exists e_t such that*

$$\mathbb{E}[\Delta^t \mid w^t] = -\eta \nabla F(w^t) + e_t, \tag{22}$$

and

$$\|e_t\| \leq C_1(\delta + \epsilon_b + \epsilon_m) + C_2 c_{\text{ortho}} + C_3 \|\hat{r}_b^t - r_b^t\|. \tag{23}$$

967 *Proof* Let $\mathcal{E}_t$ denote the good filtering event in A5. Since $\Pr(\mathcal{E}_t^c) \leq \delta$ and $\|\Delta^t\| \leq G$ by A6,

$$968 \quad \mathbb{E}[\Delta^t \mid w^t] - \mathbb{E}[\Delta^t \mid w^t, \mathcal{E}_t] \leq 2G\delta. \quad (24)$$

971 On $\mathcal{E}_t$, the total removed benign weight and retained malicious weight are bounded by
 972 ϵ_b, ϵ_m . Thus screening induces bias $O(G(\epsilon_b + \epsilon_m))$.

974 By Lemma 1, GDC distortion is bounded by $O(c_{\text{ortho}})$. Replacing r_b^t with $\hat{r}_b^t$ adds error
 975 $O(\|\hat{r}_b^t - r_b^t\|)$.

977 Collecting terms yields (22)–(23). □

980 **Theorem 3** Let $\eta \leq \frac{1}{4L}$. Then

$$982 \quad \frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E} \|\nabla F(w^t)\|^2 \leq \frac{F(w^0) - F^*}{c\eta T} + O(\eta\sigma^2) + O(\eta\zeta^2) \\
 983 \quad + O(\delta + \epsilon_b + \epsilon_m) + O(c_{\text{ortho}}^2) \\
 984 \quad + O\left(\frac{1}{T} \sum_{t=0}^{T-1} \|\hat{r}_b^t - r_b^t\|^2\right). \quad (25)$$

991 *Proof* By A1 (smoothness),

$$993 \quad F(w^{t+1}) \leq F(w^t) + \eta \langle \nabla F(w^t), \Delta^t \rangle + \frac{L\eta^2}{2} \|\Delta^t\|^2. \quad (26)$$

996 Taking expectation and using Lemma 2,

$$998 \quad \mathbb{E} \langle \nabla F(w^t), \Delta^t \rangle = -\eta \mathbb{E} \|\nabla F(w^t)\|^2 + \mathbb{E} \langle \nabla F(w^t), e_t \rangle. \quad (27)$$

$$1002 \quad \mathbb{E} \langle \nabla F(w^t), e_t \rangle \leq \frac{\eta}{2} \mathbb{E} \|\nabla F(w^t)\|^2 + \frac{1}{2\eta} \mathbb{E} \|e_t\|^2.$$

1004 Thus

$$1006 \quad \mathbb{E} F(w^{t+1}) \leq \mathbb{E} F(w^t) - \frac{\eta}{2} \mathbb{E} \|\nabla F(w^t)\|^2 \\
 1007 \quad + \frac{1}{2\eta} \mathbb{E} \|e_t\|^2 + \frac{L\eta^2}{2} \mathbb{E} \|\Delta^t\|^2. \quad (28)$$

$$\begin{aligned}\mathbb{E}\|\Delta^t\|^2 &\leq 4\eta^2\mathbb{E}\|\nabla F(w^t)\|^2 + 4\mathbb{E}\|e_t\|^2 \\ &\quad + O(\eta^2\sigma^2) + O(\eta^2\zeta^2).\end{aligned}\tag{29}$$

Substituting (29) into (28) and using $\eta \leq 1/(4L)$,

$$\begin{aligned}\mathbb{E}F(w^{t+1}) &\leq \mathbb{E}F(w^t) - c\eta\mathbb{E}\|\nabla F(w^t)\|^2 \\ &\quad + C\eta\mathbb{E}\|e_t\|^2 + C\eta^3(\sigma^2 + \zeta^2).\end{aligned}\tag{30}$$

Summing (30) from $t = 0$ to $T - 1$ telescopes:

$$\begin{aligned}c\eta \sum_{t=0}^{T-1} \mathbb{E}\|\nabla F(w^t)\|^2 &\leq F(w^0) - F^\star \\ &\quad + C\eta \sum_{t=0}^{T-1} \mathbb{E}\|e_t\|^2 + CT\eta^3(\sigma^2 + \zeta^2).\end{aligned}\tag{31}$$

Dividing by $c\eta T$ yields (25). $\square$

Corollary. Setting $\eta = \Theta(1/\sqrt{T})$ gives

$$\begin{aligned}\frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E}\|\nabla F(w^t)\|^2 &= O\left(\frac{1}{\sqrt{T}}\right) + O\left(\frac{\sigma^2}{\sqrt{T}}\right) + O\left(\frac{\zeta^2}{\sqrt{T}}\right) \\ &\quad + O(\delta + \epsilon_b + \epsilon_m) + O(c_{\text{ortho}}^2) \\ &\quad + O\left(\frac{1}{T} \sum_{t=0}^{T-1} \|\hat{r}_b^t - r_b^t\|^2\right).\end{aligned}\tag{32}$$

6 Experimental Results

This section assesses the effectiveness and robustness of FedDSG across a range of adversarial threat models and data heterogeneity conditions. We first detail the experimental configuration, followed by a presentation of the overall performance, robustness analyses, practical considerations, and ablation study results.

1059 6.1 Experimental Setup

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1061 6.1.1 FL Setting

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1063 We use SGD with learning rate 0.01 and momentum 0.9, without weight decay, and
1064 fix the batch size to 128. Local training runs 2 epochs for MNIST/FashionMNIST
1065 and 5 epochs for SVHN/CIFAR-10. We simulate a cross-device federation with 100
1066 clients in total. In each communication round, we randomly sample 50% of clients
1067 and run training for 100 rounds. This protocol matches cross-device IoT federations
1068 where only a subset of intermittently available devices participates per round.
1069

1070

1074 6.1.2 Datasets and Models

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1076 We evaluate on MNIST [37], FashionMNIST [38], SVHN [39], and CIFAR-10 [40].
1077 Unless otherwise stated, data are partitioned using Dirichlet($\alpha = 1$) to emulate het-
1078 erogeneous device-local distributions. For MNIST/FashionMNIST we use a five-layer
1079 CNN with batch normalization and dropout. For SVHN/CIFAR-10 we adopt ResNet-
1080 18. In all experiments, FedDSG uses a small server-side clean subset to form the
1081 geometric reference direction; unless otherwise specified, the anchor ratio is fixed at
1082 5% of the training set.
1083

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1089 6.1.3 Attack Settings

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1091 We employ a continuous poisoning regime throughout training. The malicious-client
1092 ratio (MCR) is adapted to dataset complexity: 0.6 for MNIST and FashionMNIST,
1093 and 0.4 for SVHN and CIFAR-10. Each malicious client poisons 40% of its local
1094 training samples according to the prescribed poisoned-data ratio.
1095

1096 We evaluate three representative classes of backdoor attacks commonly studied in
1097 federated learning:
1098

1099

1100 **Centralized Backdoor Attack (CBA).** All malicious clients embed the same
1101 global trigger into a subset of training samples and relabel them to a designated
1102

target class. This classical single-trigger scenario establishes a consistent backdoor pattern across compromised participants. The uniform trigger distribution makes CBA relatively detectable but highly effective when the malicious ratio is sufficient.

Distributed Backdoor Attack (DBA). DBA partitions the global trigger into disjoint local patterns distributed among malicious clients. Each compromised participant embeds only one partition, and the full trigger emerges through aggregation. This decomposition enhances stealth by reducing per-client anomaly signatures, particularly under heterogeneous data distributions.

Label-Flipping Attack (LFA). LFA reassigns labels of samples from a base class to an attacker-chosen target class without modifying input features. While this untargeted poisoning does not establish a trigger-dependent backdoor, the mislabeling induces measurable distortions in final-layer bias parameters. We include LFA to evaluate semantic detection mechanisms under poisoning that alters class distributions rather than input patterns.

6.1.4 Baseline Methods

We compare FedDSG with representative baselines that include robust aggregation and detection-based defenses, as well as methods that combine these two components. Specifically, we include FedAvg, Krum [9], Median [10], FoolsGold [33], FLAME [8], FLTrust [26], FedDLAD [12], and FLAB [34]. All baselines are implemented following their original configurations to ensure fair comparison.

6.1.5 Evaluation Metrics

We evaluate all methods using three standard metrics:

- **Attack Success Rate (ASR).** ASR is the fraction of triggered test samples classified into the attacker-specified target class, measuring the effectiveness of the embedded backdoor.

Table 2 Overall comparison results.

Dist.	Defense	MNIST			FMNIST			SVHN			CIFAR-10		
		ACC $\uparrow$	ASR $\downarrow$	OPS $\uparrow$	ACC $\uparrow$	ASR $\downarrow$	OPS $\uparrow$	ACC $\uparrow$	ASR $\downarrow$	OPS $\uparrow$	ACC $\uparrow$	ASR $\downarrow$	OPS $\uparrow$
D1	FedAvg	0.993	1.000	0.000	0.863	1.000	0.000	0.933	0.989	0.000	0.656	0.993	0.000
	Krum	0.809	1.000	-0.185	0.478	1.000	-0.446	0.266	1.000	-0.726	0.301	0.983	-0.531
	Median	0.988	1.000	-0.005	0.819	1.000	-0.051	0.927	0.994	-0.011	0.665	0.994	0.013
	FoolsGold	0.981	0.999	-0.011	0.785	0.990	-0.080	0.919	0.989	-0.015	0.695	0.992	0.060
	FLAME	0.980	1.000	-0.013	0.806	1.000	-0.066	0.888	0.994	-0.053	0.587	0.993	-0.105
	FLTrust	0.960	0.004	0.963	0.748	0.015	0.852	0.805	0.980	-0.128	0.535	0.984	-0.175
	FedDLAD	0.943	1.000	-0.050	0.708	1.000	-0.180	0.914	0.993	-0.024	0.584	0.995	-0.112
	FLAB	0.992	1.000	-0.001	0.844	1.000	-0.022	0.849	0.999	-0.100	0.638	0.347	0.623
	FedDSG (ours)	0.959	0.003	0.963	0.741	0.006	0.853	0.915	0.007	0.974	0.647	0.091	0.895
D2	FedAvg	0.992	1.000	0.000	0.868	1.000	0.000	0.926	0.995	0.000	0.666	0.995	0.000
	Krum	0.965	1.000	-0.027	0.741	1.000	-0.146	0.628	1.000	-0.327	0.322	1.000	-0.522
	Median	0.990	1.000	-0.002	0.840	1.000	-0.032	0.927	0.996	0.000	0.681	0.994	0.024
	FoolsGold	0.987	0.001	0.994	0.807	0.004	0.926	0.930	0.987	0.012	0.678	0.993	0.020
	FLAME	0.979	1.000	-0.013	0.805	1.000	-0.073	0.914	0.012	0.975	0.578	0.999	-0.136
	FLTrust	0.974	0.003	0.979	0.770	0.010	0.877	0.848	0.980	-0.069	0.560	0.948	-0.112
	FedDLAD	0.985	1.000	-0.007	0.795	1.000	-0.084	0.925	0.996	-0.002	0.675	0.995	0.014
	FLAB	0.992	0.001	0.999	0.885	0.002	1.018	0.921	0.047	0.947	0.661	0.126	0.866
	FedDSG (ours)	0.961	0.005	0.964	0.757	0.006	0.866	0.921	0.040	0.954	0.657	0.122	0.864

ACC denotes clean accuracy, ASR denotes attack success rate, and OPS denotes the overall performance score. Dist. denotes the distribution setting, where D1 = Non-IID and D2 = IID. The best result in each column is highlighted with underlined boldface, and the second-best result is shown in boldface; ties are allowed.

- **Accuracy (ACC).** ACC is the standard top-1 accuracy on clean test inputs and reflects the model’s utility under normal operation.

- **Overall Performance Score (OPS).** OPS summarizes the robustness–utility trade-off:

$$\text{OPS} = \frac{D_{\text{ACC}} - B_{\text{ACC}}}{B_{\text{ACC}}} - \frac{D_{\text{ASR}} - B_{\text{ASR}}}{B_{\text{ASR}}}, \quad (33)$$

where D_{ACC} and D_{ASR} denote the accuracy and ASR under the defense, and B_{ACC} and B_{ASR} correspond to the FEDAVG baseline. Higher OPS indicates better combined improvement in accuracy and attack resistance.

6.2 Overall Performance

Table 2 reports ACC, ASR, and OPS under both Non-IID and IID settings. The Non-IID regime is the primary focus, as it reflects the heterogeneous and skewed data distributions characteristic of IoT deployments. In this setting, existing defenses display clear instability: FLAB suppresses the attack only on CIFAR-10, FedDLAD leaves consistently high ASR, and FLTrust performs well on MNIST but degrades sharply on more complex datasets such as CIFAR-10 and SVHN. These inconsistencies lead to large variance in OPS across tasks.

FedDSG maintains a notably stable profile under Non-IID conditions. Across all four datasets, ASR remains below 0.10 while ACC stays near the benign reference, producing the highest OPS in every case. This consistency is particularly evident on SVHN and CIFAR-10, where benign updates exhibit large natural dispersion and most baselines either lose accuracy or fail to neutralize the backdoor influence. The results indicate that FedDSG is well aligned with the challenges of IoT-scale heterogeneity.

The IID results serve as a supporting sanity check. When client distributions become aligned, the performance gap across defenses narrows, yet FedDSG continues to deliver low ASR and competitive accuracy. Although FLAB and FoolsGold achieve

Table 3 Attack-wise performance on CIFAR-10.

Defense	CBA			DBA			LFA
	ACC $\uparrow$	ASR $\downarrow$	OPS $\uparrow$	ACC $\uparrow$	ASR $\downarrow$	OPS $\uparrow$	ACC $\uparrow$
FedAvg	0.656	0.993	0.000	0.669	0.989	0.000	0.606
Krum	0.301	0.983	-0.531	0.183	0.989	-0.726	0.253
Median	0.665	0.994	0.013	0.644	0.993	-0.041	0.598
FoolsGold	0.695	0.992	0.060	0.678	0.989	0.013	0.608
FLAME	0.587	0.993	-0.105	0.521	0.998	-0.230	0.545
FLTrust	0.535	0.984	-0.175	0.530	0.088	0.703	0.513
FedDLAD	0.584	0.995	-0.112	0.543	0.994	-0.193	0.529
FLAB	0.638	0.347	0.623	0.647	0.118	0.848	0.650
FedDSG(ours)	0.647	0.091	0.895	0.662	0.103	0.885	0.644

Results are reported on CIFAR-10 under Non-IID partitioning for CBA, DBA, and LFA attacks. ACC denotes clean accuracy, ASR denotes attack success rate, and OPS denotes the overall performance score.

strong results on select datasets, their behavior remains more dataset-dependent. FedDSG preserves a balanced profile across all tasks, confirming that its advantages under Non-IID conditions do not come at the cost of degraded performance in the IID regime.

6.3 Robustness Analysis

6.3.1 Different Attacks

Table 3 summarizes the results on CIFAR-10 under three attack settings. FedDSG attains the lowest ASR and the highest OPS for both CBA and DBA, while keeping ACC comparable to the stronger baselines. FLAB performs well on LFA, and FedDSG reaches a similar accuracy level, indicating that it does not penalize clean-label scenarios. Across all three attack types, other defenses either leave the attack largely intact or incur noticeable drops in ACC. FedDSG remains the only method that performs consistently across all settings.

6.3.2 Heterogeneous Poisoning Intensity

This experiment evaluates whether FedDSG remains stable when the malicious score distribution is deliberately dispersed. The reviewer-raised scaling-factor manipulation

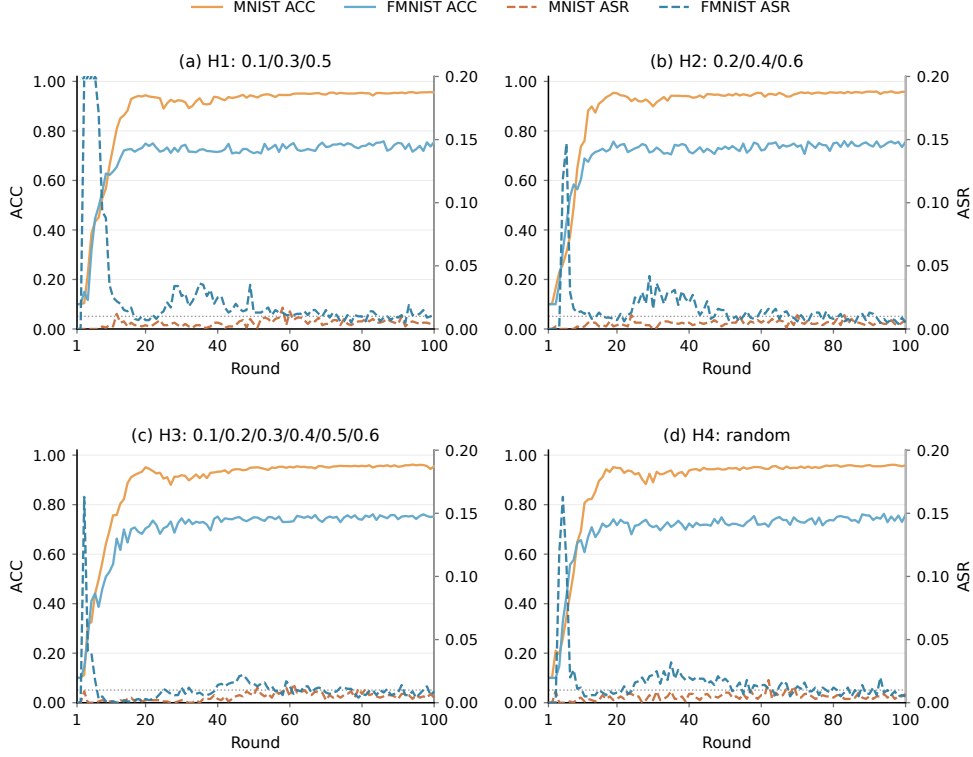


Fig. 3 Training curves under score-smearing attack simulation. H1, H2, H3, and H4 denote malicious clients using poisoned-data ratios of 0.1/0.3/0.5, 0.2/0.4/0.6, 0.1/0.2/0.3/0.4/0.5/0.6, and random ratios, respectively. These heterogeneous attack intensities are used to disperse the malicious SBF score distribution.

aims to smear the one-dimensional SBF scores and weaken the separability of k -means. For SBF, however, simple global scaling of a bias update has limited influence on the score because the score is constructed from concentration and dominance ratios of class-wise bias energies, which mainly describe the relative shape of the bias pattern rather than its absolute magnitude. Therefore, we use heterogeneous poisoned-data ratios as a practical stress test for dispersed malicious score distributions by assigning different poisoned-data ratios to malicious clients. This setting changes the strength of target-class reinforcement across malicious updates and produces a less compact malicious score distribution, making the SBF clustering step more challenging.

Fig. 3 reports the training curves on MNIST and FMNIST under four heterogeneous poisoning configurations. Across all settings, ASR may briefly increase in the early rounds but rapidly decreases and remains low afterwards. Meanwhile, ACC converges to a stable level. These results indicate that FedDSG does not rely on an ideal two-cluster separation of SBF scores. When malicious scores are dispersed, SBF still removes high-salience malicious updates, while GDC further constrains residual drift from updates that are not perfectly filtered. This confirms that the SBF–GDC pipeline remains robust under score-smearing attack intensities.

6.3.3 Impact of MCR

Fig. 4. (a) and Fig. 4. (b) show that FedAvg fails across the entire sweep, with ASR remaining close to one. FedDSG keeps ACC stable and ASR low as MCR increases. On CIFAR-10, ASR stays below 0.15 even at higher MCR values, and ACC changes only slightly. Several baselines degrade more noticeably, either retaining many malicious updates or losing clean accuracy once the proportion of malicious clients becomes moderate to high.

6.3.4 Impact of PDR

Fig. 4. (c) and Fig. 4. (d) examine the effect of increasing poisoning intensity within malicious clients. FedAvg becomes increasingly vulnerable as PDR rises, whereas FedDSG maintains low ASR throughout. On CIFAR-10 and SVHN, ASR for multiple baselines rises sharply once PDR exceeds 0.3, and ACC drops accordingly. FedDSG shows only mild ACC variation and consistently lower ASR across the entire range.

6.3.5 Impact of Non-IID Severity (Dirichlet α)

Fig. 4. (e) and Fig. 4. (f) vary the Dirichlet parameter to adjust the level of heterogeneity. Most baselines deteriorate as α decreases, especially on CIFAR-10 and SVHN, where benign updates become widely dispersed. FedDSG shows only small changes

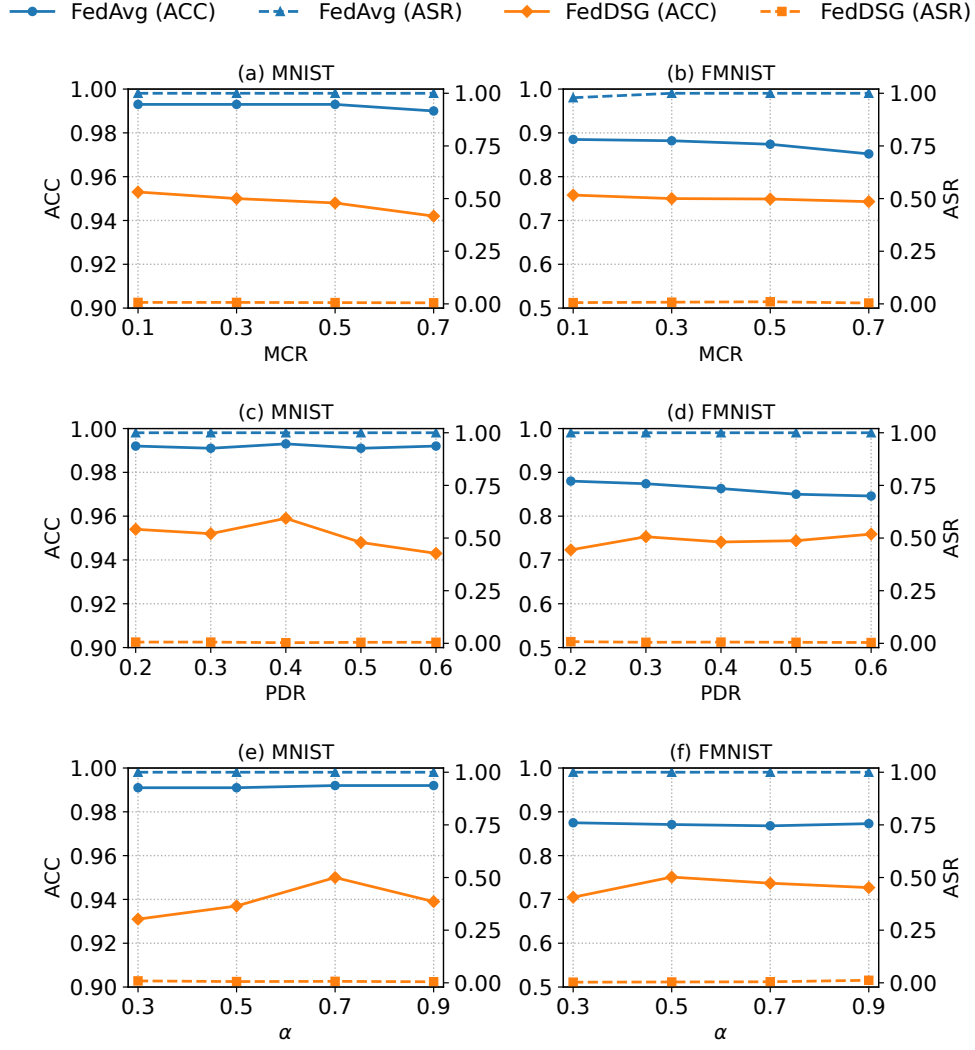


Fig. 4 Sensitivity analysis. Panels (a)–(b) vary MCR, (c)–(d) vary PDR, and (e)–(f) vary Dirichlet α .

across the entire range. Accuracy remains close to the benign reference, and ASR stays below ten percent even at strong heterogeneity. The overall trend indicates that FedDSG is less sensitive to shifts in benign update dispersion than the competing methods.

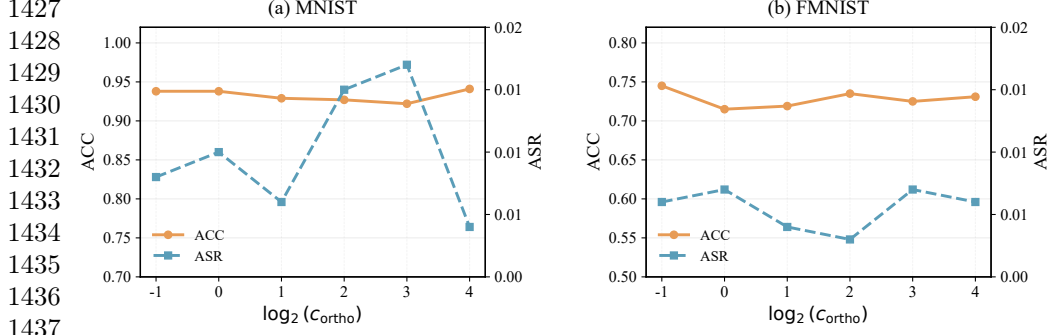


Fig. 5 Sensitivity to the GDC cone-width parameter c_{ortho} . The horizontal axis uses $\log_2(c_{\text{ortho}})$ to cover a wide range of geometric constraint strengths.

6.3.6 Impact of c_{ortho}

This experiment studies the sensitivity of FedDSG to the cone-width parameter c_{ortho} . In GDC, the retained bias update is decomposed into a reference-aligned component $b_s^{\parallel}$ and an orthogonal component $b_s^{\perp}$, and the constraint $\|b_s^{\perp}\| \leq c_{\text{ortho}} \|b_s^{\parallel}\|$ limits the maximum deviation from the anchor-induced reference direction. Equivalently, if θ_s denotes the angle between the client bias update and the reference direction, c_{ortho} controls the upper bound of $\tan(\theta_s)$. Fig. 5 shows that FedDSG maintains stable ACC and low ASR over a relatively wide range of c_{ortho} values on MNIST and FMNIST. Smaller values impose stronger geometric constraints, whereas larger values preserve more directional diversity among retained updates. The results indicate that FedDSG is not dependent on a narrowly tuned c_{ortho} and can maintain a stable trade-off between benign update diversity and residual backdoor suppression.

6.4 Practical Considerations

6.4.1 Anchor Budget Sensitivity

Fig. 6 evaluates anchor budgets between 0.5% and 5%. Performance remains largely stable throughout this range. Accuracy varies by less than three percentage points, and ASR stays below one percent on all datasets. FMNIST shows slightly larger

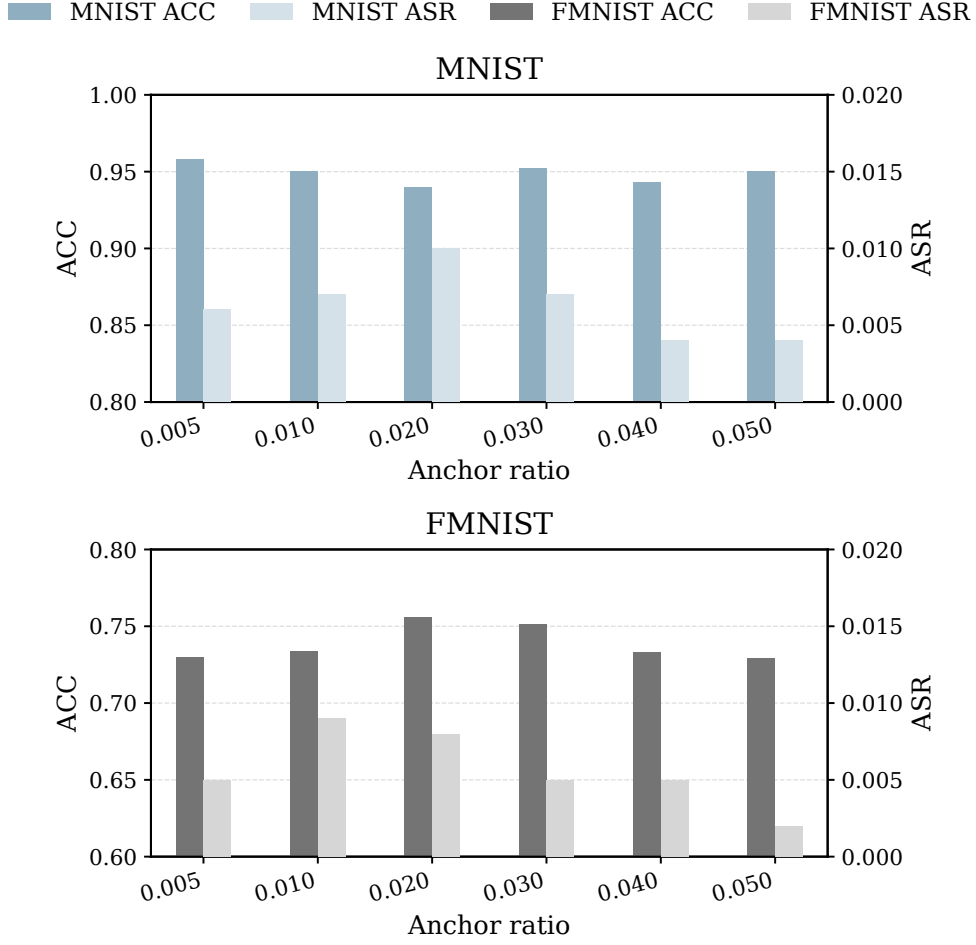


Fig. 6 Anchor-ratio sensitivity.

fluctuations, but no consistent trend with respect to anchor size. These results suggest that FedDSG can operate effectively with very small trusted-data budgets, making it practical for IoT deployments with limited server-side resources.

6.4.2 Anchor-quality Sensitivity

This experiment examines whether the anchor set must be strictly class-balanced. We evaluate two degraded anchor conditions: Dirichlet-imbalanced anchors and missing-class anchors where one class, three classes, or the target class are removed from

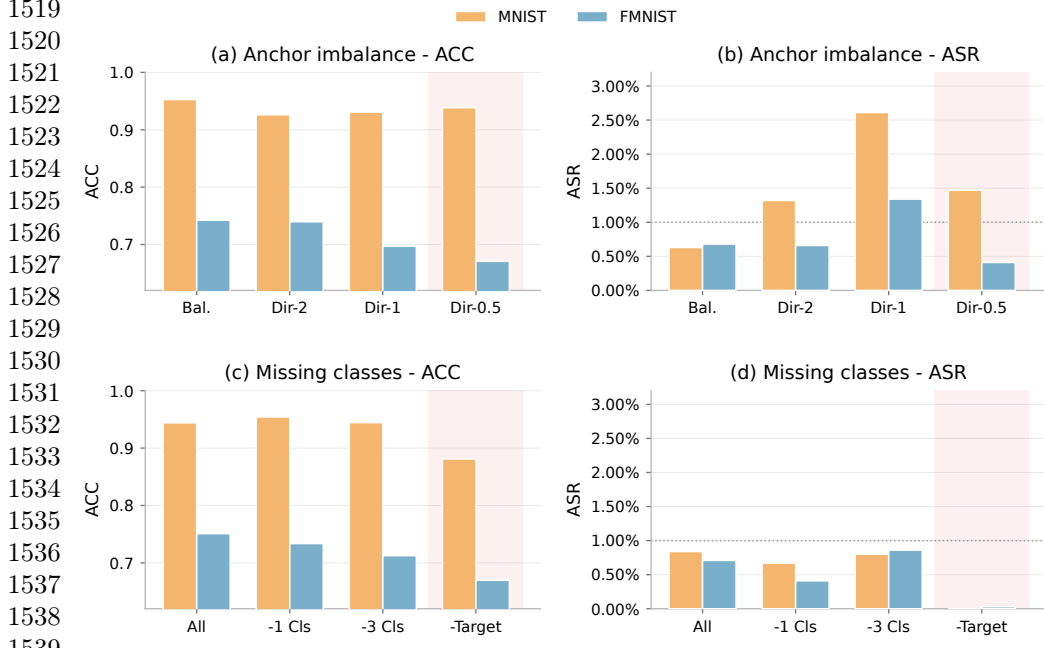


Fig. 7 Anchor-quality ablation. Panels (a)–(b) evaluate imbalanced anchor sets, and panels (c)–(d) evaluate anchors with missing classes. “Bal.” denotes a balanced anchor, “Dir” denotes Dirichlet-imbalanced anchors, and “-Target” denotes removing the target class from the anchor set.

the anchor set. As shown in Fig. 7, FedDSG remains effective under both types of anchor degradation. Under anchor imbalance, ACC remains relatively stable and ASR stays low in most cases. Under missing-class settings, FedDSG continues to suppress the backdoor effectively, including the case where the target class is absent from the anchor set. This indicates that GDC does not rely on memorizing target-class anchor samples. Instead, the anchor set mainly provides a coarse clean reference direction for constraining directional drift. Severe anchor degradation can still affect clean accuracy, especially on FMNIST, but the backdoor suppression remains stable.

6.4.3 Runtime Efficiency

Fig. 8 summarizes end-to-end cost under identical training configurations on a server with one NVIDIA RTX 5090 GPU (32 GB), 25 vCPUs (Intel Xeon Platinum 8470Q),

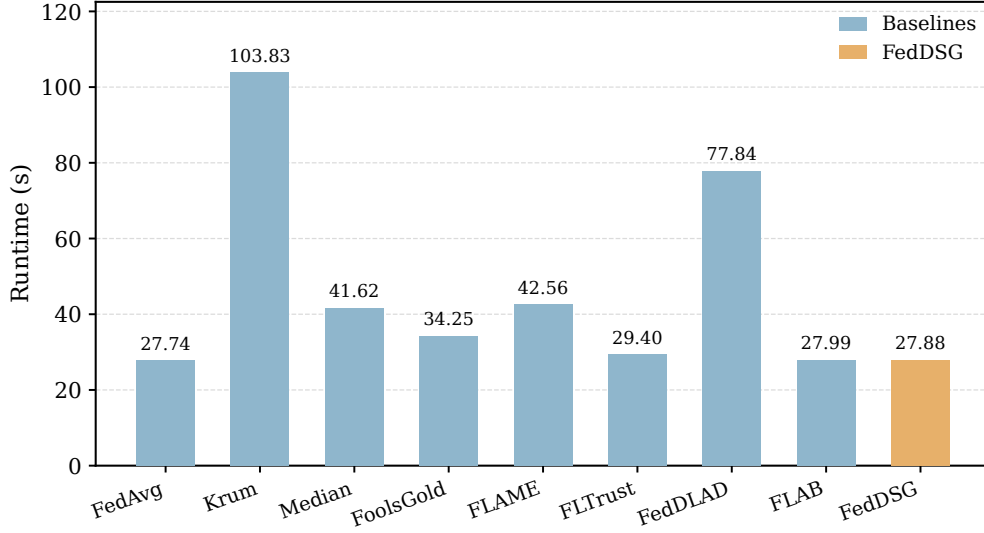


Fig. 8 Defense runtime comparison.

90 GB RAM, PyTorch 2.0.0, and CUDA 11.8. Krum and FedDLAD incur substantial overhead due to pairwise distance computations, clustering, or iterative filtering. FedDSG remains close to FedAvg by processing only the last-layer bias block with single-pass vector operations. Per-round processing requires at most $O(|S_t|K)$ and avoids full-dimensional pairwise computation, confirming negligible additional cost.

6.5 Ablation Study

Table 4 reports the performance of SBF-only, GDC-only, and the full FedDSG system under IID and Non-IID partitions.

SBF-only. SBF reliably identifies the target class and removes a portion of malicious clients under IID data. Under Non-IID settings, however, benign bias updates spread more unevenly across classes, and the score distribution of benign and malicious clients begins to overlap. Several attackers therefore remain after filtering, and their combined effect is enough to keep ASR at a high level. The core issue is that a

Table 4 Module ablation results.

Data	Module		MNIST		FMNIST		CIFAR-10		SVHN	
	SBF	GDC	ACC	ASR	ACC	ASR	ACC	ASR	ACC	ASR
IID	✓		0.992	0.001	0.889	0.002	0.668	0.090	0.922	0.870
		✓	0.992	1.000	0.871	1.000	0.669	0.995	0.931	0.983
	✓	✓	0.961	0.005	0.757	0.006	0.657	0.122	0.921	0.040
Non-IID	✓		0.992	1.000	0.849	1.000	0.640	0.128	0.925	0.941
		✓	0.990	1.000	0.847	1.000	0.672	0.993	0.926	0.986
	✓	✓	0.959	0.003	0.741	0.006	0.647	0.091	0.915	0.007

SBF denotes the semantic bias filter and GDC denotes the geometric direction constraint. ACC denotes clean accuracy and ASR denotes attack success rate.

semantic score alone cannot separate clients well once benign class preferences vary widely.

GDC-only. GDC preserves clean accuracy because it keeps updates close to a common server-side direction. But without a mechanism to detect adversarial behavior, a continuous attacker can adjust its update to stay roughly aligned with this direction while still pushing the model toward the target class. Accuracy remains high, yet ASR remains close to the undefended case. GDC controls the geometry of updates, but not whether an update carries malicious intent.

Full FedDSG. Using both modules removes the weaknesses observed above. SBF filters out the clients that exhibit clear target-class amplification, and GDC stabilizes the remaining updates so that small adversarial shifts are no longer effective. As a result, FedDSG achieves low ASR and maintains accuracy across all datasets, including the more difficult Non-IID settings where both semantic overlap and geometric variability are present. Neither module alone reaches this level of robustness.

The full pipeline removes dominant target-class amplification and limits orthogonal drift after filtering. This complementary structure explains why only FedDSG consistently suppresses ASR while preserving accuracy across regimes.

7 Limitations

FedDSG has practical limitations that should be acknowledged. The method is mainly designed for backdoor attacks whose malicious optimization leaves observable target class reinforcement in the bias of the final layer. This assumption is consistent with common data poisoning and local objective manipulation attacks. However, fully adaptive attackers with white box knowledge may deliberately suppress evidence in the bias parameters, for example by freezing the bias of the final layer or shifting the backdoor effect to weight parameters, which remains outside the main threat model and may weaken SBF. In addition, GDC relies on a small clean anchor set to estimate a benign reference direction. Although the anchor quality experiments show that FedDSG does not require an anchor set that is strictly balanced across classes and remains effective under imbalanced or partially incomplete anchor data, obtaining trusted calibration samples at the server may still be challenging in privacy sensitive or highly specialized IoT deployments. These limitations suggest that future work should extend FedDSG toward semantic constraints across multiple layers, constraints on weight parameters, reference estimation without anchors, and privacy preserving anchor construction.

8 Conclusion

We propose FedDSG, a server-side defense for federated learning that remains robust under heterogeneous data and elevated malicious participation. FedDSG combines a semantic bias filter (SBF) and a geometric direction constraint (GDC). The SBF is based on a novel scale-invariant semantic cue extracted from last-layer bias updates, while the GDC is guided by a small anchor reference. Together, they use semantic information to detect abnormal target-class reinforcement and apply geometric correction to constrain directional drift, forming a complementary defense against backdoor manipulation. Extensive experiments demonstrate that FedDSG consistently reduces attack success rates while maintaining competitive accuracy across diverse attack

settings. The method also operates entirely at the server without modifying client protocols and incurs only modest computational overhead. Future work will explore adaptive reference estimation to reduce reliance on fixed anchor data and enhance robustness against more advanced adversarial strategies.

Supplementary information. Not applicable.

List of abbreviations

Abbreviation	Definition
ACC	Clean accuracy
ASR	Attack success rate
CBA	Centralized Backdoor Attack
DBA	Distributed backdoor attack
EMA	Exponential moving average
FL	Federated learning
GDC	Geometric direction constraint
IID	Independent and identically distributed
IoT	Internet of Things
LFA	Label-flipping attack
MCR	Malicious client ratio
OPS	Overall performance score
PDR	Poisoned data ratio
SBF	Semantic bias filter
SVHN	Street View House Numbers

Declarations

Availability of data and materials

The datasets used in this study are publicly available benchmark datasets, including MNIST, Fashion-MNIST, CIFAR-10, and SVHN, from their respective official repositories.

The source code and configuration files supporting the findings of this study are publicly available in the FedDSG repository: <https://github.com/shenmiren197/>

FedDSG. All experiments can be reproduced using the provided code, configuration files, and publicly available datasets.

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